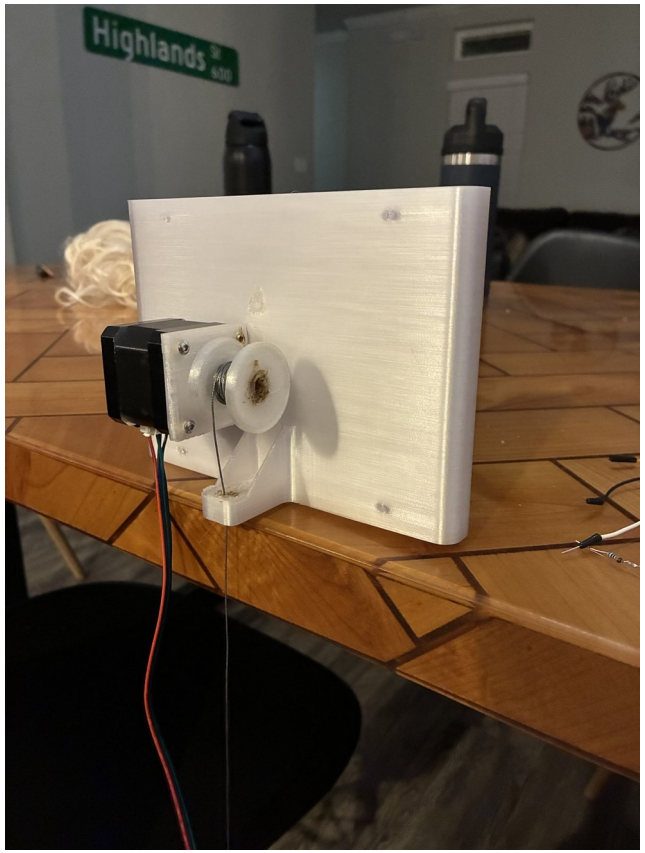
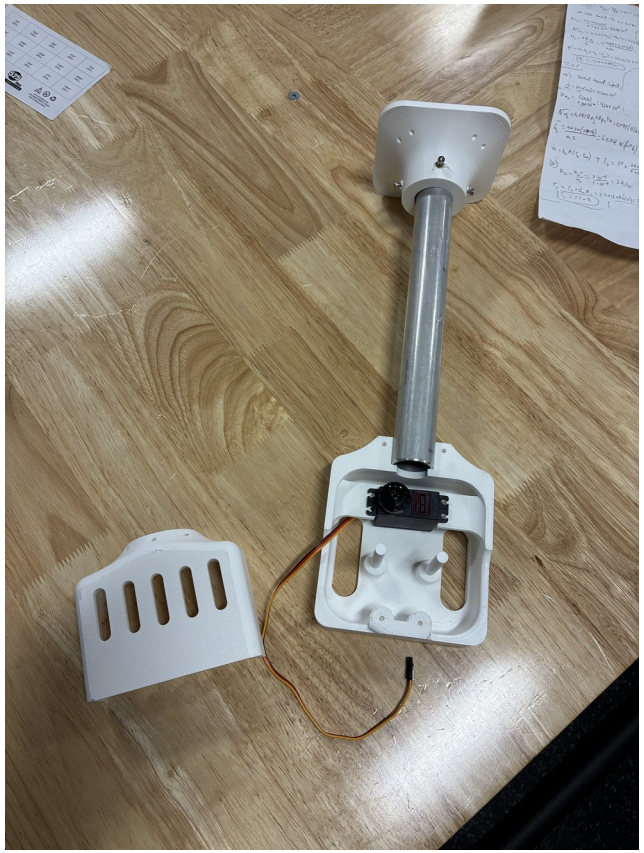
RoomCleaner

A cable-driven robot that picks laundry up off the floor

Jasper Buntinx
Mechanical Engineering, Texas A&M University
October 27, 2026





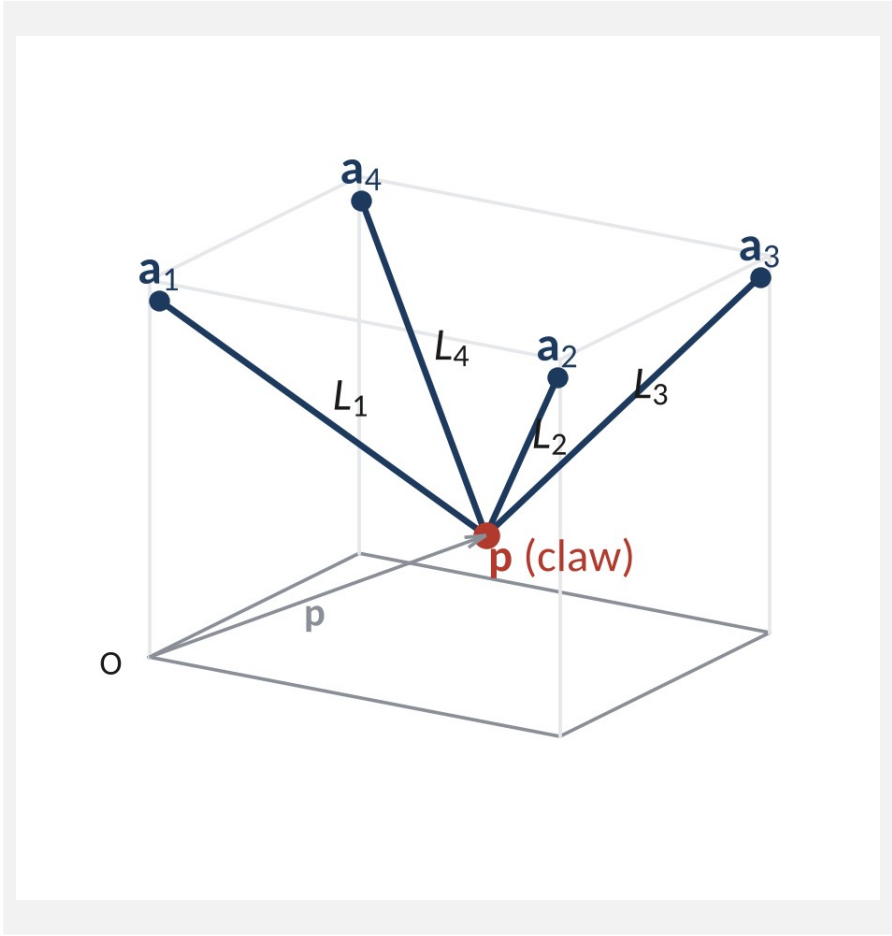
About Me

- B.S. Mechanical Engineering, Texas A&M University, *[graduation month and year]*
- Built EcoFil, a filament recycler with a shredder, dryer and extruder
- Metal sculptures for the Austin Creek Show, made with fiber laser cutting and welding
- Outside of class: *[hobbies and interests]*
- Why this project: *[one sentence]*

[Photo of you building: EcoFil, a sculpture, or the workshop]

Agenda

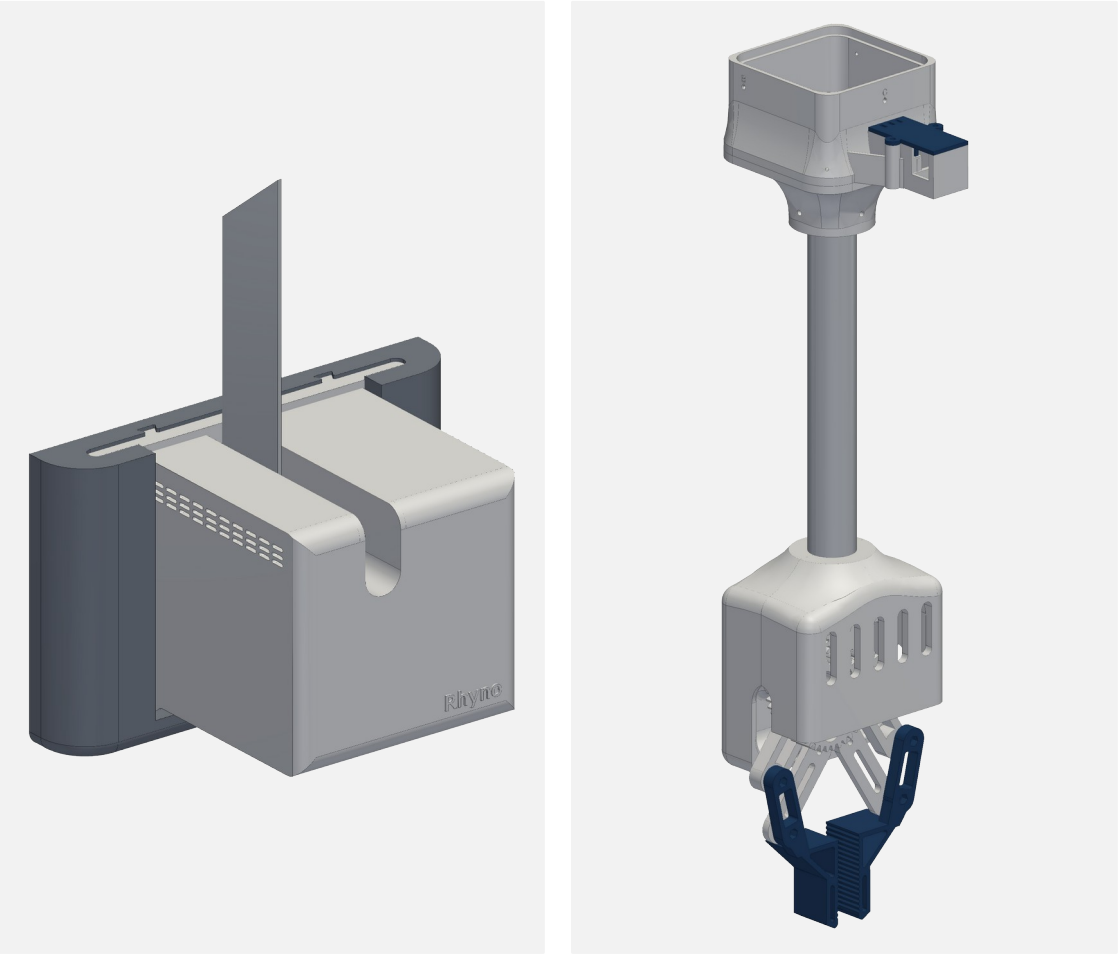
1. Project goal, scope and timeline
2. Design: targets, trade-offs and key calculations
3. Build: printed parts, machining and electronics
4. Test: first system test and thermal verification
5. Good and poor decisions, and lessons learned



Cables from the ceiling anchors a_1 to a_4 to the claw at p

Project Goal

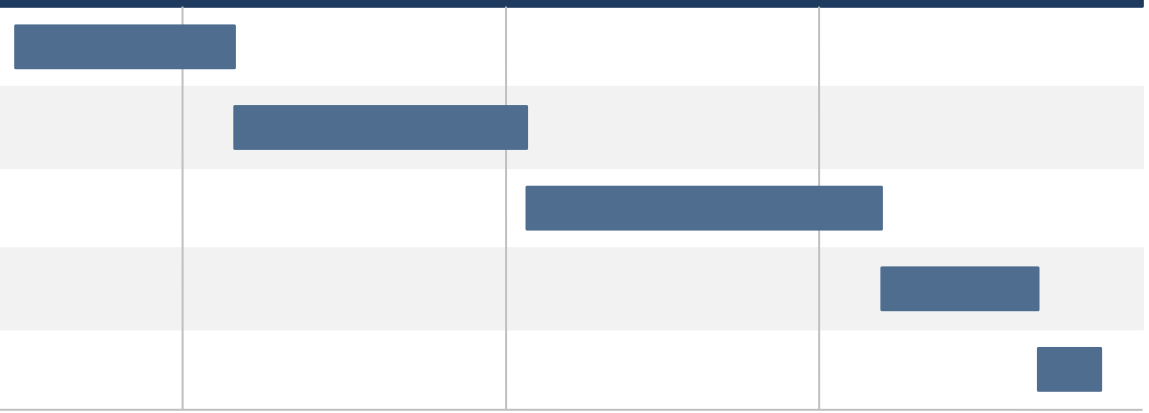
- Pick up laundry anywhere in an 11 × 9 ft room and drop it in a hamper
- Four ceiling-mounted winches suspend one claw on cables
- Design load: jeans, 0.9 kg, at 2 m/s² acceleration; first test load is 1 lb
- 3D-printed PETG (polyethylene terephthalate glycol) parts on a hobby budget
- Success: one garment delivered to the hamper with no human input



Final design (CAD): wall station, left; claw, right

Scope and Timeline

Phase	Dates	July	August	September	October
Concept and requirements	Jul 16 – Aug 5				
Generation 1: build and bench test	Aug 6 – Sep 2				
Redesign: sensors and wiring	Sep 3 – Oct 6				
Generation 2 and final design	Oct 7 – Oct 21				
Build and first system test	Oct 22 – Oct 27				



My work (solo project)

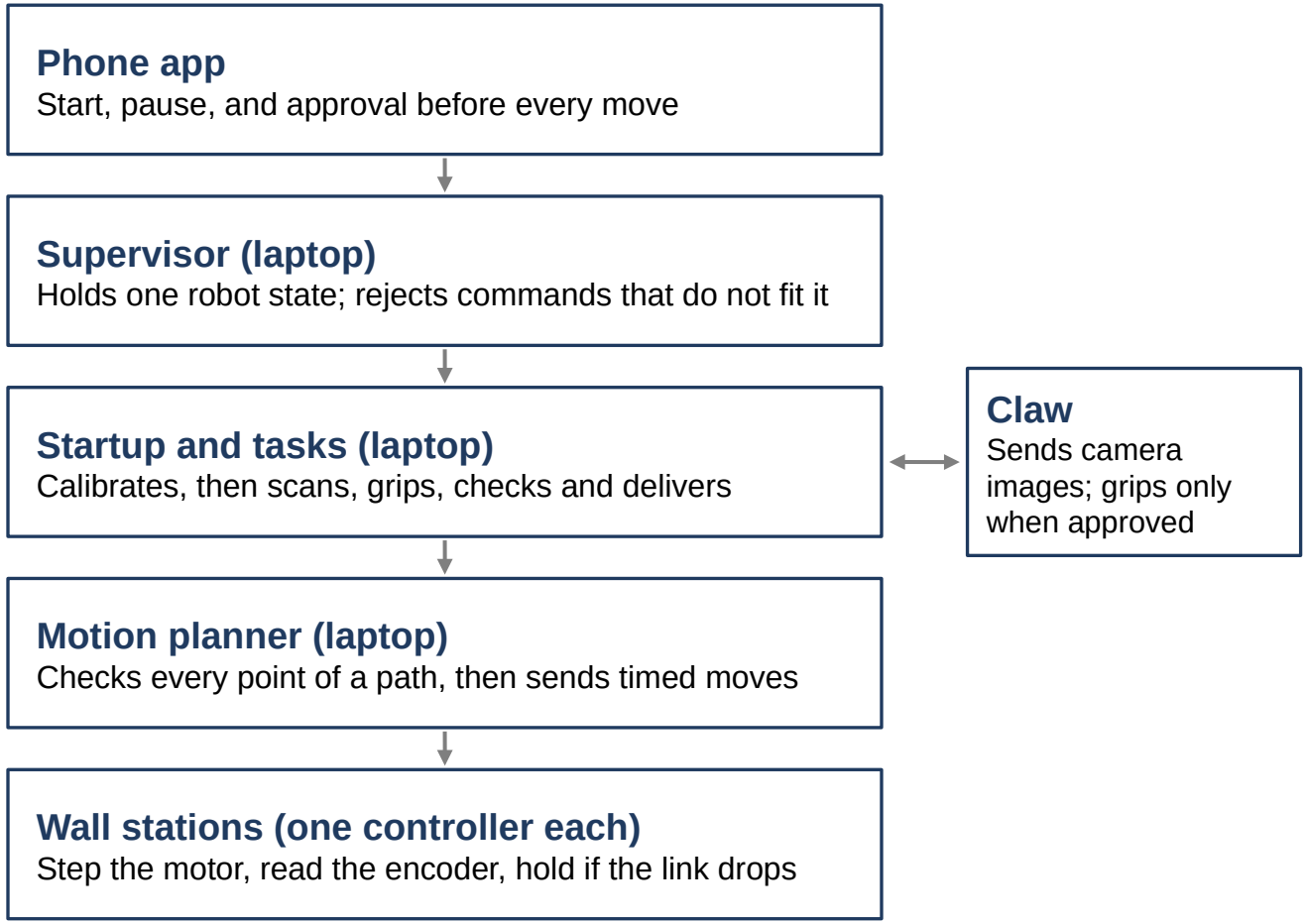
- Requirements, calculations and test plans
- All CAD (computer-aided design) in SolidWorks
- 3D printing, fit testing and machining the heat fin
- Wiring, soldering and electronics bring-up
- Control software: *[confirm the parts you wrote]*

Purchased or external

- Stepper motors, drivers and controller boards
- Servo, camera, battery and encoder boards
- Garment detection: YOLO-World, an open-source model
- Shop access or mentors: *[if any]*

How It Works

- Four wall stations, each with its own winch, encoder and controller
- One claw with a gripper, camera and battery, hung on four cables
- Laptop plans each move and checks every cable tension before it runs
- Phone app starts, pauses and approves each move
- Software simulated end to end; hardware test this week



Commands flow down; status and camera images flow up.

Design Targets

Parameter	Target	Current value	Status
Lift capacity	1.76 kg: claw, battery and jeans	1.56 kg (calculated)	**Not met**
Claw mass	0.45 kg	0.86 kg with battery (estimated); *[weighed]*	**Not met**
Motor temperature	Below 60 °C (PETG limit)	58 °C prototype fin; 45–49 °C with 1/8 in fin	Marginal
Working area	Full 11 × 9 ft floor	All but 0.2 m along the walls	Met in design
Wiring	One cable per station	One 12 V pair; 0.51 V drop on the longest run	Met
Positioning accuracy	2 mm per cable	0.7 mm (calculated)	To be tested
Speed	50 mm/s at 2 m/s²	Not yet run	To be tested
Cost	Hobby budget	*[total cost]*	*[fill in]*

Safety is a fixed constraint and is not traded against any of these targets.

Design Trade-offs

Design change	Benefit	Cost	Chosen setting
Higher motor current	41 % more pull at 1.5 A	Heat rises with current squared: 2.0 W to 4.05 W	1.5 A, vented shell and heat fin
Stronger claw	Gear safety factor from 0.4 to 2.0	More claw mass, less payload	Module 1.5 gears
Larger room	More floor covered	Longer wires, more voltage drop	Only 12 V power runs to the stations
Working closer to the ceiling	More working volume	Cable tension rises from 12 N to 32 N	Stay 0.4 m below the anchors
Smaller spool radius	More pull per amp	Slower line speed	15 mm core: 34 N empty, 25 N full
Larger safety margins	No skipped steps or wall contact	Less usable force and floor area	Force ÷ 1.5; 0.2 m from walls

Key coupling: a heavier claw needs more pull, more pull needs more current, and heat rises with the square of the current.

Cable Tension

- Cables can only pull, so tension must stay positive
- Tension rises sharply near the ceiling
- 32 N per cable at 0.15 m below the anchors; 8.3 N at 0.6 m
- Operating limit: 0.4 m below the anchors

Cable tension at the operating limit

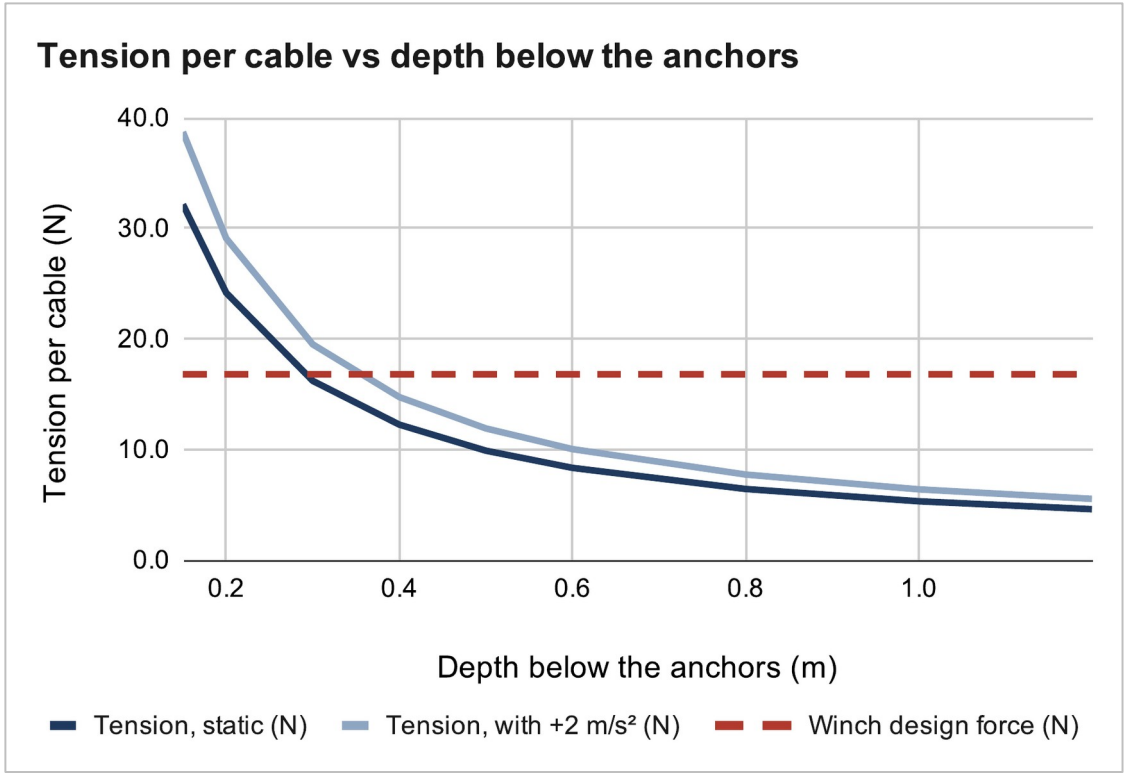
$$T = W / (4 \sin \varphi), \quad \sin \varphi = d / \sqrt{(d^2 + D_h{}^2)}$$

$$\sin \varphi = 0.4 / \sqrt{(0.4^2 + 2.17^2)} = 0.181$$

$$T = 8.87\ \text{N} / (4 \times 0.181) = \mathbf{12.2\ N}$$

$$\text{With } 2\ \text{m/s}^2\text{: } 12.2 \times 1.20 = \mathbf{14.7\ N} < 16.8\ \text{N}$$

Where: W = (0.45 + 0.454) kg × 9.81 m/s² = 8.87 N; d = 0.4 m below the anchors; D_h = 2.17 m, room center to corner; 16.8 N = winch design force



Source: RoomCleaner engineering calculations, “Cable tension” (calculated values)

Winch Pull

- NEMA 17 (National Electrical Manufacturers Association) stepper with a printed spool
- Driver current raised to the rated 1.5 A (was 1.06 A)
- Pull: 34 N with an empty spool, 25 N when full
- Luggage-scale pull test to confirm the torque

Winch design force at 1.5 A

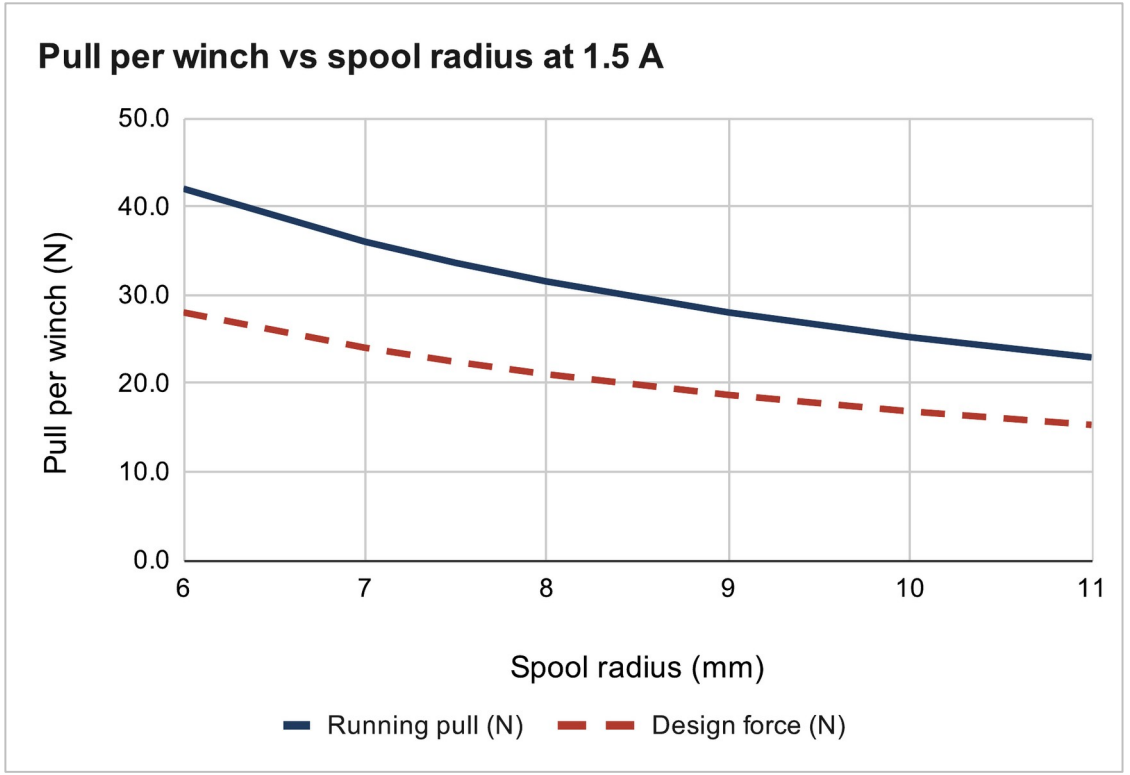
$$I = V_{ref} / (8\,R_s) = 1.2\ V / (8 \times 0.1\ \Omega) = \mathbf{1.5\ A}$$

$$\tau_{run} = 0.7 \times 0.36\ N{\cdot}m = 0.252\ N{\cdot}m$$

$$T = \tau / r = 0.252\ N{\cdot}m / 0.010\ m = \mathbf{25.2\ N}$$

$$F_{design} = 25.2\ N / 1.5 = \mathbf{16.8\ N}$$

Where: V_{ref} = driver reference voltage; R_s = sense resistor; 0.36 N·m = holding torque at 1.5 A; 0.7 = running/holding ratio (assumed); r = 10 mm full spool; 1.5 = skip margin



Source: RoomCleaner engineering calculations, “Winch pull” (calculated values)

One Controller per Winch

- Generation 1 drove all four motors from one central board
- Adding sensors at each winch required 10 wires per corner
- Moved to a separate controller in each station
- Total wire reduced from 820 ft to 164 ft

Wire length and timing error

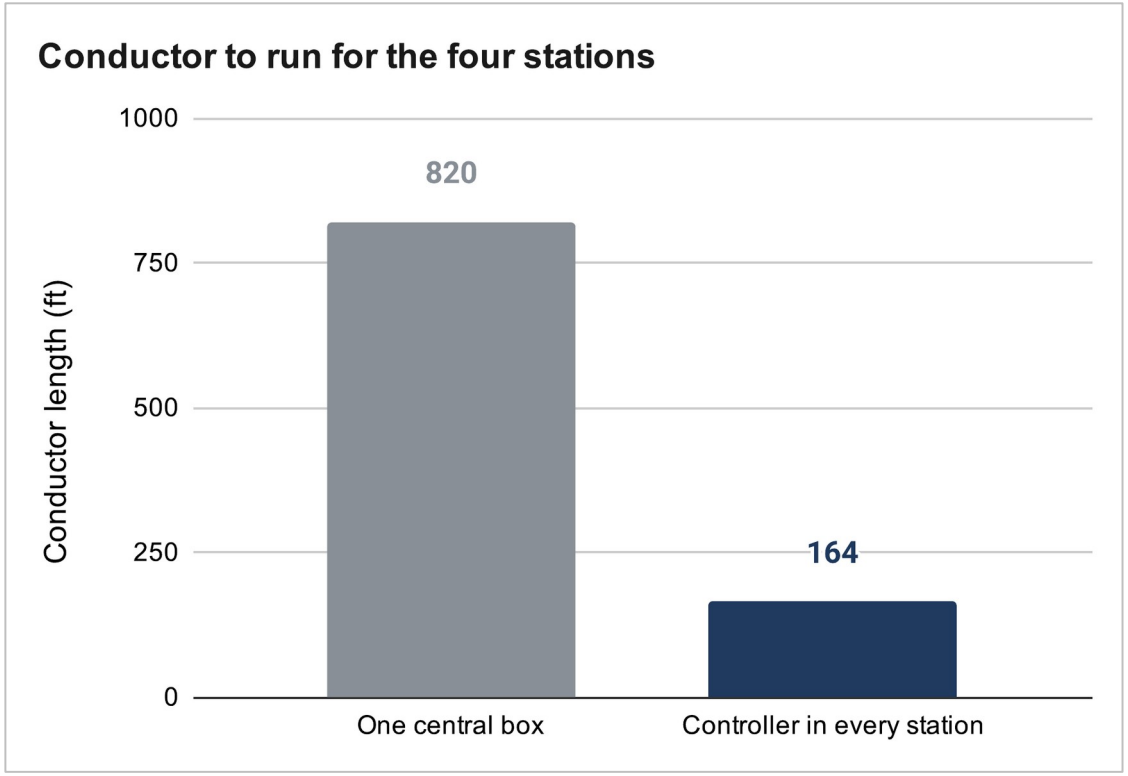
Wire = $n \times \Sigma L$

Central board: 10×82 ft = **820 ft**

Controller per station: 2×82 ft = **164 ft**

Timing error: $\Delta x = v\,\Delta t = 50\ \text{mm/s} \times 50\ \text{ms}$ = **2.5 mm**

Where: n = conductors per corner (motor 4, switch 2, encoder 4); ΣL = 6 + 21 + 23 + 32 ft of cable routes; v = line speed; Δt = clock offset between stations



Source: RoomCleaner engineering calculations, “Wiring” (calculated values)

Motor Cooling

- Sealed PETG shell traps motor heat
- PETG creeps above 60 °C under load
- Aluminum fin added through the top of the shell
- 90-minute hold test at 1.5 A planned

Motor heat and temperature at 1.5 A

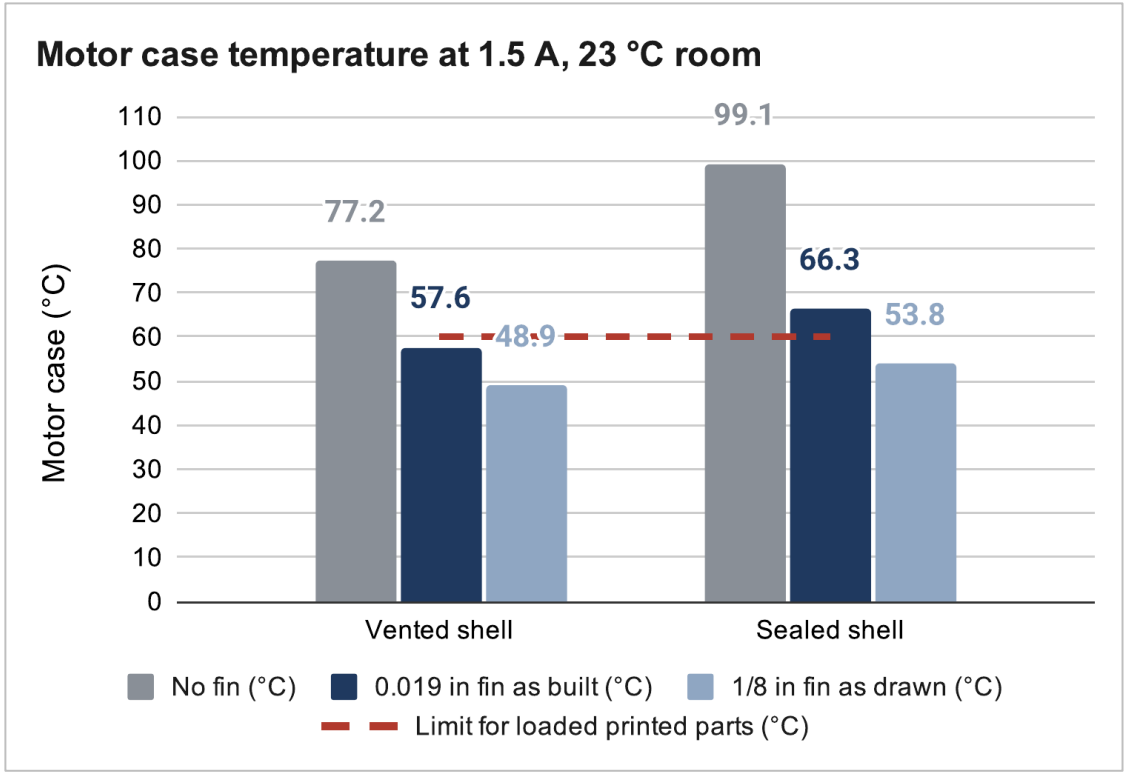
$$P = I^2 R_{coil} = (1.5\ A)^2 \times 1.8\ \Omega = \mathbf{4.05\ W}$$

$$1/R_{th} = 1/R_{shell} + 1/R_{fin},\quad \Delta T = P \times R_{th}$$

$$0.019\text{ in fin: } R_{fin} = 23.1\ K/W \rightarrow \mathbf{57.6\ °C}$$

$$1/8\text{ in fin: } R_{fin} = 11.9\ K/W \rightarrow \mathbf{48.9\ °C}$$

Where: R_{coil} = 1.8 Ω per phase (assumed); R_{shell} = 12 + 1 K/W, motor to shell air to room; room at 23 °C; 1.5 W of electronics in the shell; limit 60 °C



Source: RoomCleaner engineering calculations, “Motor temperature” (calculated values)

Heat Fin Sizing

- 0.019 in prototype fin reaches 95 % of its capacity by 136 mm
- 1/8 in design fin is 74 % efficient at 200 mm
- Built length of 200 mm, limited by ceiling clearance

Fin length and efficiency

$m = \sqrt{(2h / (k\,t))}$, $L_{95} = 1.83 / m$

0.019 in: $m = \sqrt{(2 \times 7 / (160 \times 0.00048))} = 13.5\ /m$

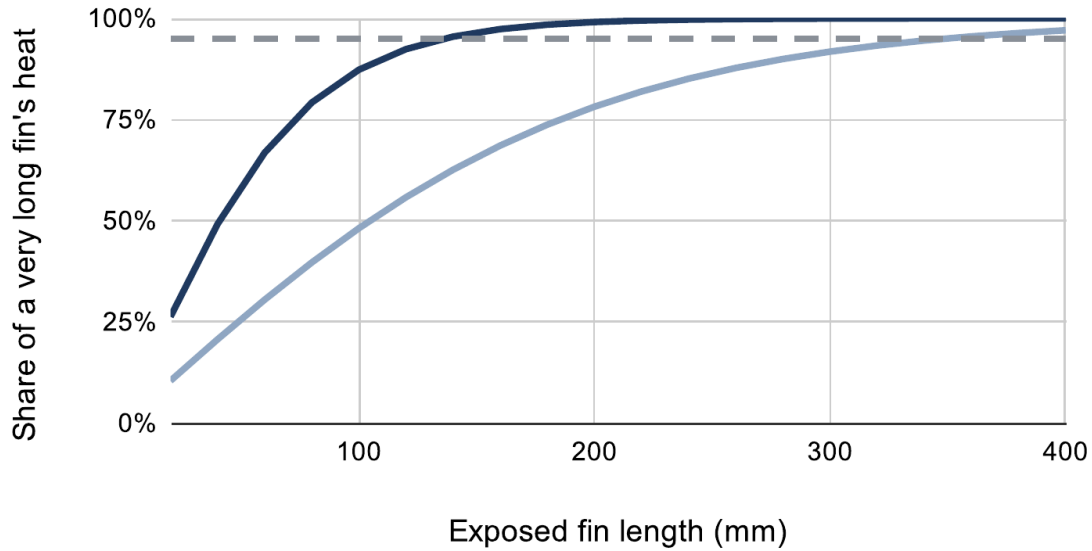
$L_{95} = 1.83 / 13.5 =$ **136 mm**

1/8 in: $m = 5.25\ /m$, $L_{95} =$ **349 mm**

$\eta = \tanh(mL) / (mL) = \tanh(1.05) / 1.05 =$ **0.74**

$R_{fin} = 1 / (7 \times 0.74 \times 0.0176) =$ **10.9 K/W**

Where: h = 7 W/m²·K, still air; k = 160 W/m·K, aluminum; t = fin thickness; L = 200 mm; A = 2 × 44 mm × 200 mm = 0.0176 m²; $R_{fin} = 1 / (h\,\eta\,A)$



Source: RoomCleaner engineering calculations, “Heat fin” (calculated values)

Homing Method

- Original plan: a bead on each cable trips a switch at the outlet
- Cables could not get shorter than their homing length
- Replaced with camera calibration on a printed floor board
- Full room now reachable; beads kept as end stops

Reachable area with bead homing

$$L_i(\mathbf{p}) \geq L_{i,\text{home}} = \|\mathbf{a}_i - \mathbf{p}_{\text{home}}\|$$

Home 1 m above the floor: **5 %** reachable

Home 2 m above the floor: **29 %** reachable

Where: a_i = anchor i; p = claw position; reachable = every cable at or above its home length, inside the 0.2 m wall and 0.4 m ceiling margins



Source: RoomCleaner engineering calculations, “Bead homing” (calculated values)

Claw Design

- Generation 1 used five TPU (thermoplastic polyurethane) fingers
- Fingers could not pinch thin fabric and were 4.6 times over the mass budget
- Final design: geared four-bar with parallel jaws and a 53 mm opening

Grip force and gear stress at servo stall

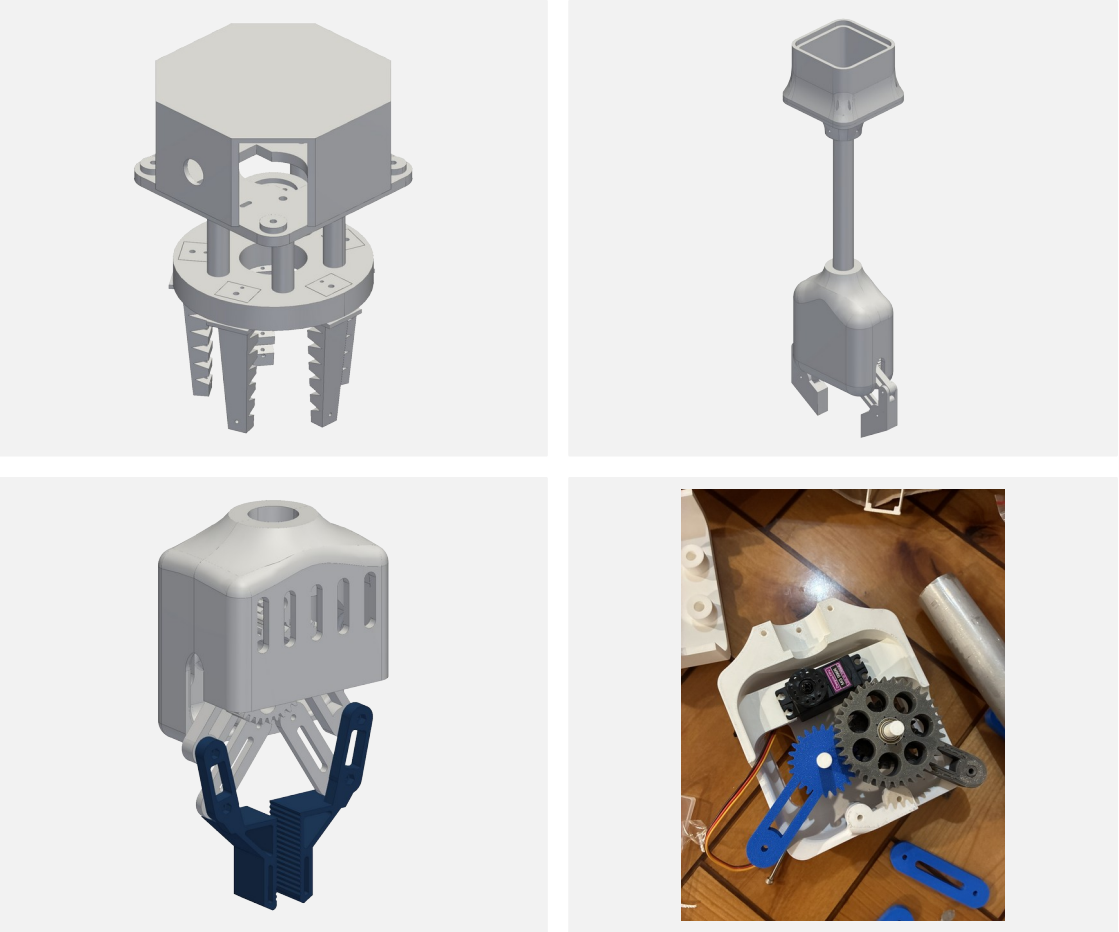
$$N \geq W / (2\mu) = 4.45\ \text{N} / 0.6 = \mathbf{7.4\ N}$$

$$F_t = 2\tau / d = 2 \times 1.078 / 0.027 = 79.9\ \text{N}$$

$$\sigma = F_t / (b\ m\ Y) = 79.9 / (15.7 \times 1.5 \times 0.309)$$

$$\sigma = 11.0\ \text{MPa},\ SF = 22.5 / 11.0 = \mathbf{2.05}$$

Where: N = grip force per jaw; W = 1 lb test load; μ = 0.3; τ = 1.078 N·m servo stall torque; d = 27 mm pitch diameter; b = 15.7 mm face width; m = 1.5 mm module; Y = 0.309 Lewis factor; SF = safety factor



Left to right, top to bottom: generation 1 fingers, v6 clamp, final four-bar (CAD), printed gear train

Weight Budget: Open Issue

- Lift capacity at 1.5 A is 1.56 kg
- Claw with battery estimated at 0.86 kg *[weighed mass]*
- 1 lb test load fits; jeans exceed capacity by 0.2 kg
- Options: a lighter claw or a 3:1 gear reduction on the winch

Lift capacity against the required mass

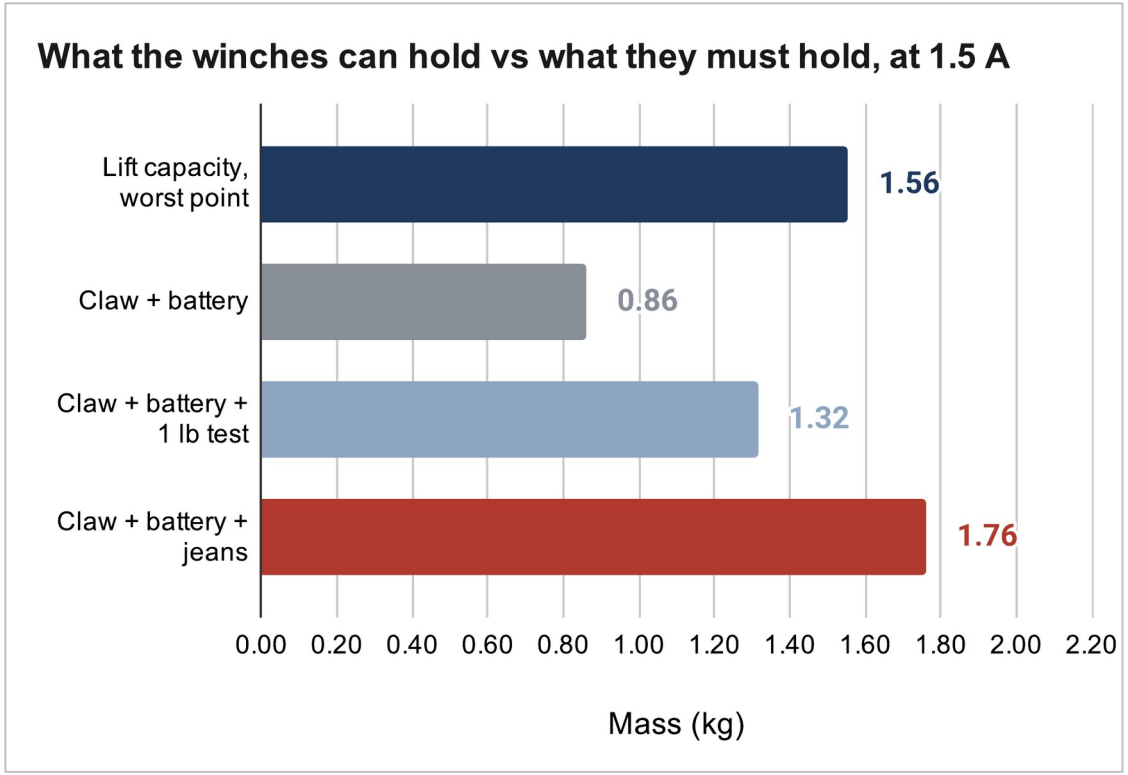
$$m_{max} = F_{design} / (k\,g)$$

$$= 16.8\ N / (1.101 \times 9.81\ m/s^2) = \mathbf{1.56\ kg}$$

$$m_{claw} = 0.55 \times 0.927 + 0.198 + 0.055 = \mathbf{0.76\ kg}$$

$$\text{Required: } 0.76 + 0.10 + 0.90 = \mathbf{1.76\ kg} > 1.56\ kg$$

Where: k = worst cable tension per unit weight over the room, with 2 m/s²; 0.55 = print fill; 0.927 kg = solid PETG from CAD; 0.198 kg tube; 0.055 kg servo; 0.10 kg battery; 0.90 kg jeans



Source: RoomCleaner engineering calculations, “Lift budget” (calculated values)

Printed Part Fits

- PETG printed on a Creality K1 at 0.2 mm layers
- Printed holes come out undersize by at least 0.2 mm
- The spool needed three prints to fit the motor shaft
- Fit coupons are now printed before full parts

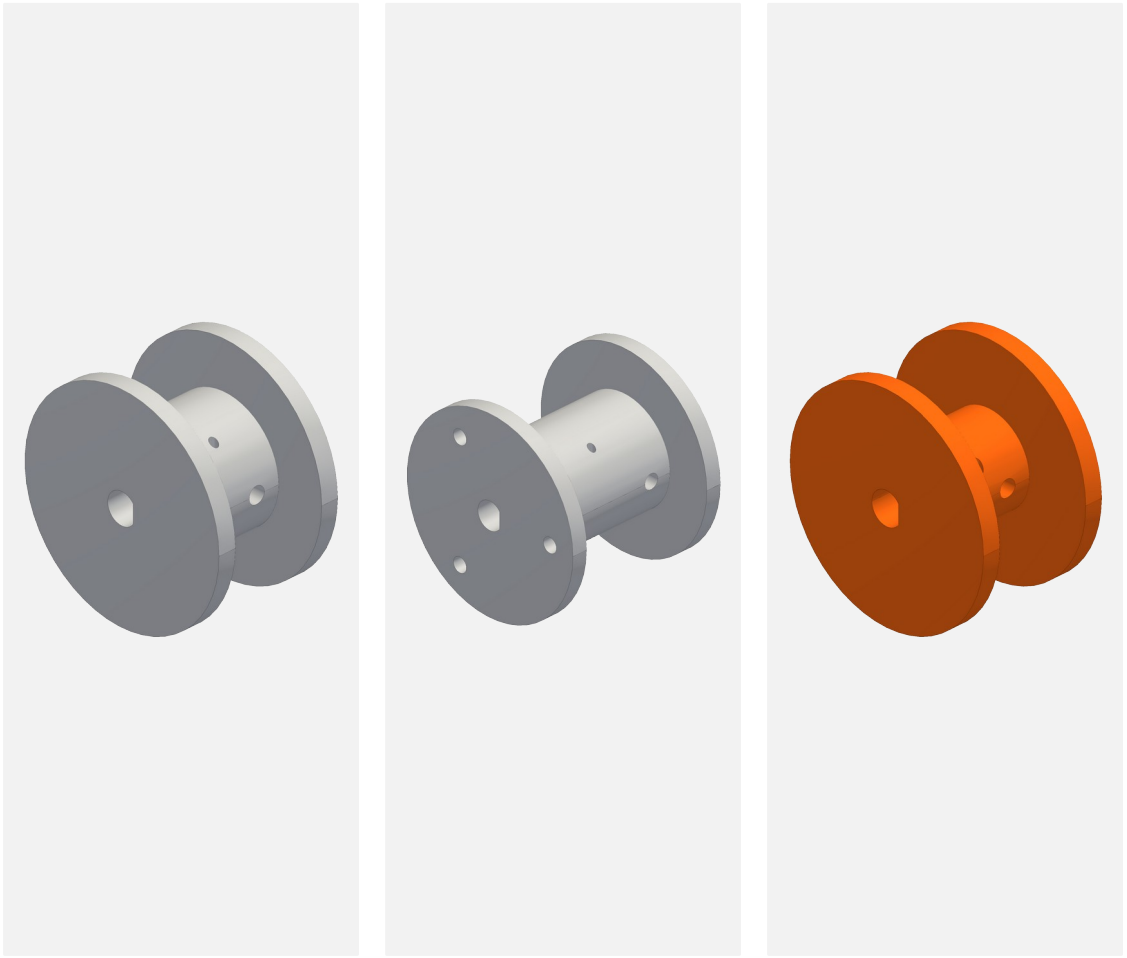
Spool bore clearance

$$c = (D_{hole} - u) - D_{shaft}$$

Ø5.2 bore too tight, so $u \geq 0.2$ mm

$c = (5.45 - 0.2) - 5.00 =$ **0.25 mm**

Where: c = clearance; u = printer undersize; D_{shaft} = 5.00 mm motor shaft; version 3 bore Ø5.45 with a D-flat for the set screw



Spool versions 1, 2 and 3 (CAD)

Heat Fin Manufacturing

- Shop drawing printed 1:1 and used as a hole template
- Prototype cut from 0.019 in aluminum sheet already on hand
- All five holes drilled in one mill setup to ±0.1 mm
- Fin bolted to the motor without reaming





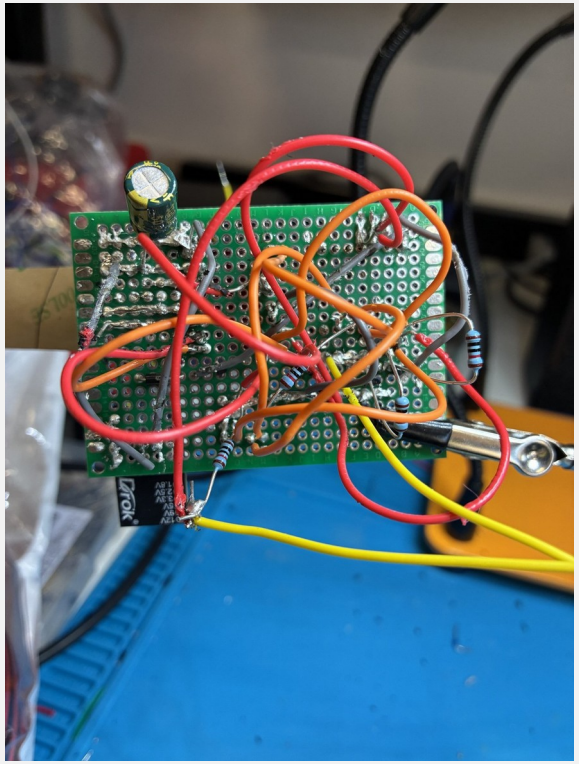




Shear and blanks, mill setup with the drawing, drilling, and the finished fin

Electronics Bring-Up

- Driver stays disabled at boot, so a reset cannot move the motor
- Schematic generated from the soldered board, 16 nets
- First motor failed to turn: logic rail measured 1.14 V instead of 3.3 V
- Supply rails are now checked with a meter before any code is loaded





Soldered station board, left; bench bring-up, right

First System Test

- Three stations: the fewest cables that fully constrain the claw
- Sequence: self-check, calibration, short move, then pickup
- 1 lb test load first, then one garment
- Pass limits defined before the run

[Photo or video from the three-station test]

Check	Pass limit	Result
Startup self-check	All stations answer; slack taken up	*[fill in]*
Calibration fit	Held-out poses inside the safety margin	*[fill in]*
Short move, 1 lb	300 mm at 50 mm/s, ≤ 2 mm per cable	*[fill in]*
Pickup	One garment into the hamper	*[fill in]*

Thermal Verification

- PETG mount must stay below 60 °C
- Hand calculation: 56–59 °C with the prototype fin, 45–49 °C with the 1/8 in fin
- SolidWorks thermal study of the motor, fin, screws and mount
- 90-minute hold test at 1.5 A to confirm

[SolidWorks temperature plot: motor, fin, screws and mount]

Method	0.019 in fin	1/8 in fin	PETG mount
Hand calculation	56–59 °C	45–49 °C	2 K below motor
SolidWorks	*[fill in]*	*[fill in]*	*[fill in]*
Hold test	*[fill in]*	*[fill in]*	*[fill in]*

Motor temperature at 1.5 A, 23 °C room, vented shell

Good and Poor Decisions

Good decision	Outcome
One controller per winch	Five times less wire; each station tested on its own
Slide-in wall mount	A station comes off while its anchor stays in place
Camera calibration instead of bead homing	Whole room reachable; no tape measure needed
Gear strength checked at servo stall	Module 1 gears would have failed; fixed in CAD before printing

Poor decision	Outcome
Mass budget set late	Claw and jeans are 0.2 kg over capacity, even at 1.5 A
Bead homing as the first plan	Under a third of the room reachable; redesign needed
Spool printed before a fit test	Three prints to fit the motor shaft
Several stations built before one was proven	Every fix repeated on each station *[confirm]*

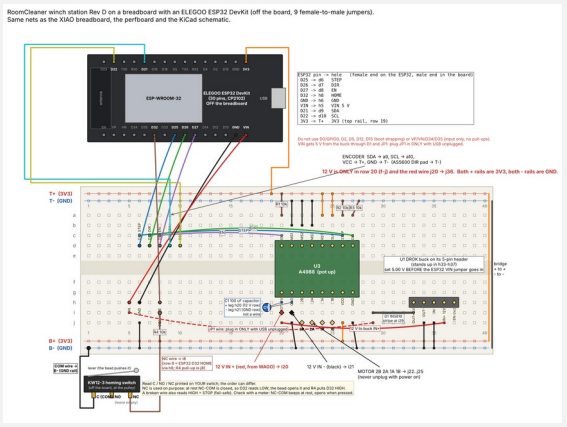
Thorough vs. Scrappy

Thorough



- Heat fin sized by calculation and machined in one mill setup to ±0.1 mm
- Station schematic checked net by net against the soldered board
- Gear strength checked at servo stall before printing

Scrappy



- Failed controller replaced by an ESP32 dev board on nine jumpers, same firmware
- Floor board made from paper tags on a 250 mm grid
- Prototype fin cut from 0.019 in sheet already on hand

Approach: thorough where a mistake is expensive or unsafe, scrappy where it is cheap to undo.

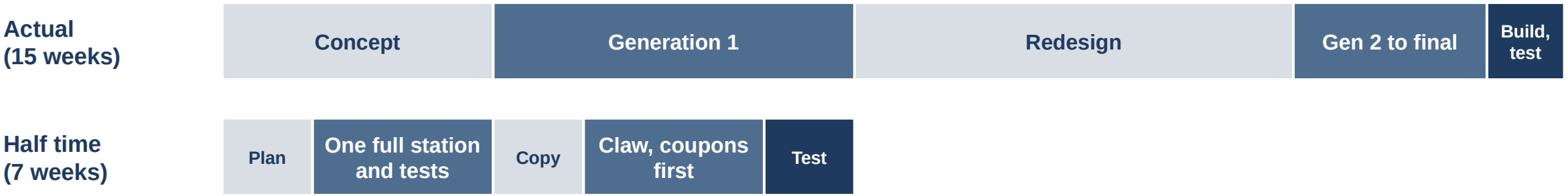
Lessons Learned

What I would do differently

- Set the mass budget early, from the measured winch pull
- Print fit coupons before full parts
- Prove one complete station before building the rest
- Check supply rails with a meter before debugging code

Completing it in half the time

- Freeze requirements in the first week
- Order long-lead parts on day one
- Start with three stations
- Limit each part to one generation, gated by a test



Same time scale. Actual schedule: July 16 to October 27.

Questions

Jasper Buntinx | jasper_buntinx@tamu.edu

Appendix

Supporting detail for questions

Project Status

Part	Status	Next test
Wall stations	Built; bench-commissioned *[how many]*	Three-station test, then the fourth
Station electronics	Built; drivers at 1.5 A *[how: Vref reading]*	90-minute hold test at 1.5 A
Heat fin	Prototype 0.019 in built; 1/8 in designed	Machine the 1/8 in fin; hold test
Thermal model of the station	Hand calculation at 1.5 A; SolidWorks model set up	Run it; then the hold test
Claw body: tube, mount, servo, gears	Built	Weigh it
Claw hands, camera housing, encoder casing	Checked in CAD; printing	Fit checks on the first print
Calibration and control software	Simulated end to end	First hardware run
Winch pull	Calculated at 1.5 A: 25–34 N	Luggage-scale pull test

Fin Sizing for 4 W

- Goal: the fin alone carries all 4.05 W
- 0.019 in sheet cannot at any length; maximum about 2 W
- 1/8 in plate needs 145–185 mm
- Thickness matters more than length

Most heat any fin length can carry

$$q_\infty = w\sqrt{(2\,h\,k\,t)} \times \theta_b$$

$$0.019\text{ in: } 0.044 \times \sqrt{(2 \times 9.3 \times 167 \times 0.000483)} \times 37$$

$$= \mathbf{2.0\ W} < 4.05\ \text{W}$$

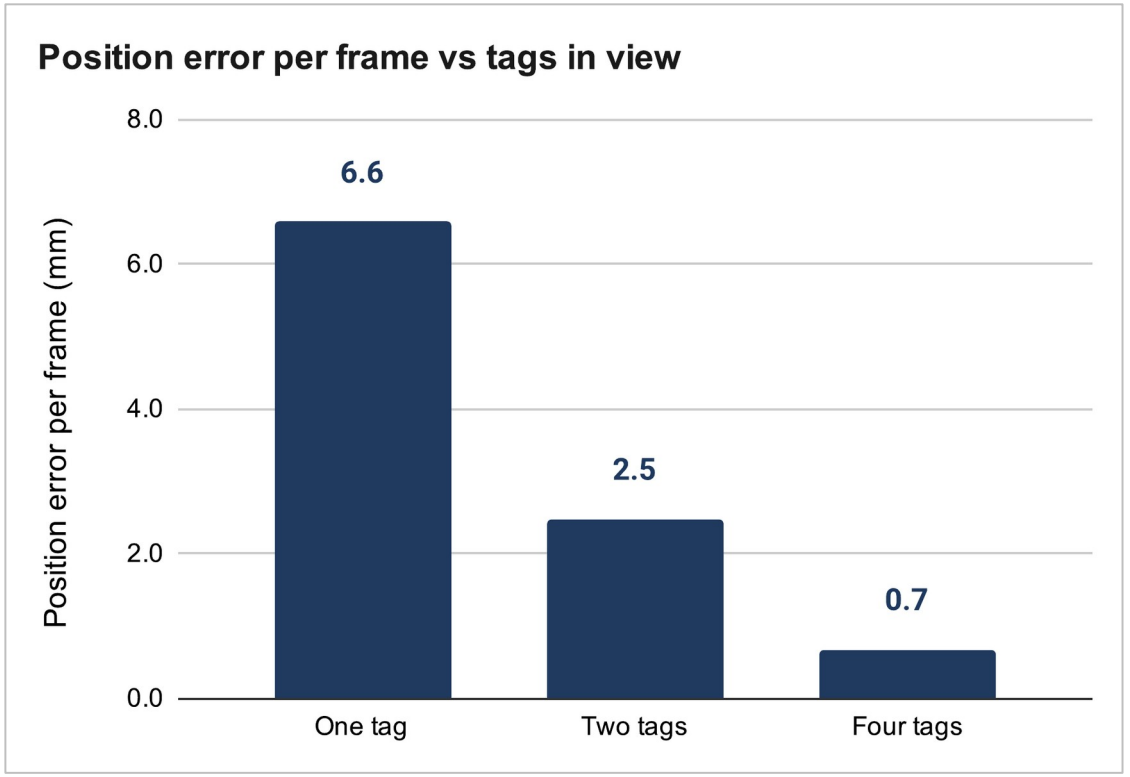
Where: w = 0.044 m fin width; h = 9.3 W/m²·K, painted; k = 167 W/m·K, 6061 aluminum; t = 0.000483 m; θ_b = 60 − 23 = 37 K

Painted 6061, 44 mm wide	Mass	Motor at 200 mm, fin alone	Shortest for ≤ 60 °C
0.019 in (prototype)	14 g	100–120 °C	None (2.0 W max)
1/16 in	47 g	66–72 °C	None (3.6 W max)
0.090 in	68 g	60–65 °C	210 mm or none
1/8 in (design)	94 g	56–59 °C	145–185 mm
3/16 in	141 g	53–55 °C	125–140 mm
1/4 in	189 g	51–52 °C	115–125 mm

All 4.05 W through the fin, 23 °C room, paste contact. Calculated.

Floor Board Accuracy

- Position error equals angle error times camera height
- One tag in view: 6.6 mm; four tags: 0.7 mm
- Averaging removes noise, but not bias



Source: RoomCleaner engineering calculations, “Floor board” (calculated values)

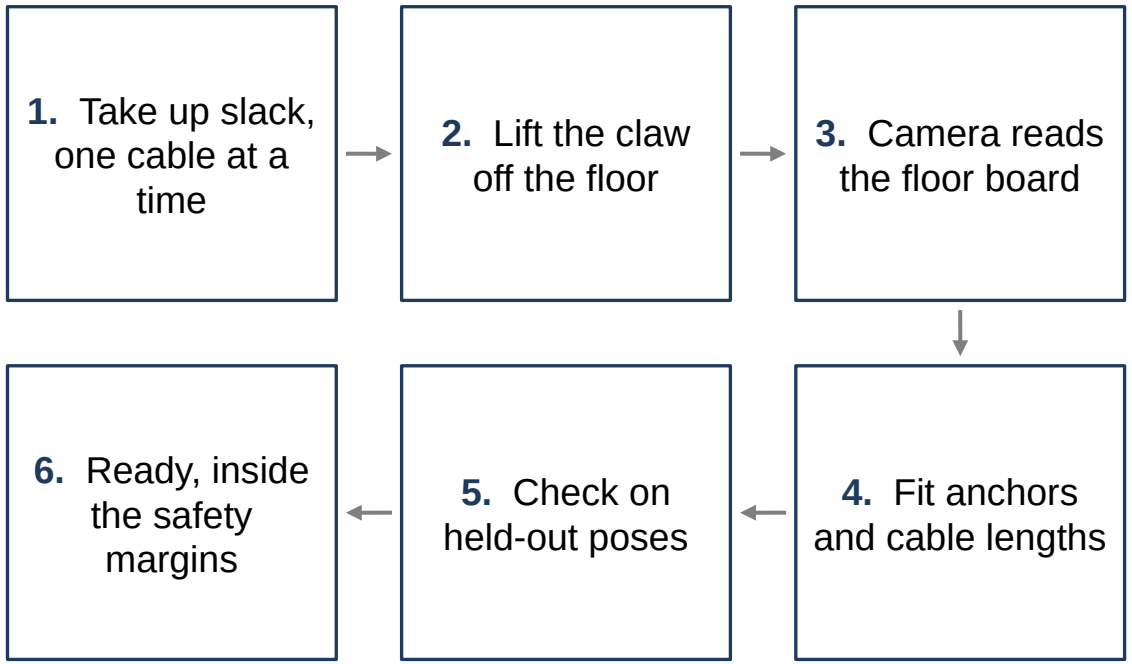
Position error with one tag in view

$$\delta x = \sigma_\theta \times L = 12.9\ \text{mrad} \times 0.51\ \text{m} = \mathbf{6.6\ mm}$$

Where: σ_θ = pose-angle noise with one tag in view; L = 0.51 m camera height above the board

Startup Calibration

- No tape measure needed
- Fits anchor positions and cable zero lengths
- Held-out poses check the fit
- Full room allowed only within the fit's error bars



Least-squares calibration fit

$$\min \Sigma_k \Sigma_i \left(L_{i0} + d_{ik} - \|\mathbf{a}_i - \mathbf{p}_k\| \right)^2$$

$$\text{Full room only if } k\,\sigma \le \text{safety margin}$$

Where: a_i = anchor i; p_k = claw pose k from the camera; d_{ik} = cable paid out, from the encoder; L_{i0} = cable zero length; σ = fit error; k = confidence multiplier

Control Logic: Generation 1 vs. Final

	Generation 1	Final
Where is the claw?	Assumed: where the cables were told to put it	Measured: claw camera on the floor board, refit at every startup
Room geometry	Point mass; anchors from a tape measure	Rigid claw; anchors and cable lengths fitted, with error bars
Who runs the motors	One central board steps all four	Each station runs time-stamped moves; the laptop coordinates
Finding laundry	Ceiling camera, pixel to floor map	Claw camera, ray from the claw's pose to the floor
Did the grab work?	Assumed once the claw closed	Camera checks pickup and delivery
Faults	Emergency stop only	Supervisor: a stop keeps torque; nothing restarts by itself

Operating Sequence

1. Start up

- Self-check
- Take up slack
- Lift a few centimeters
- Calibrate
- Set the trust level

→

2. Find

- Rise to scan height
- Detector lists the items
- Map each to the floor
- Choose the nearest

→

3. Pick and deliver

- Hover until still
- Descend and close
- Camera confirms the grip
- Move to the hamper and release
- Camera confirms the drop

→

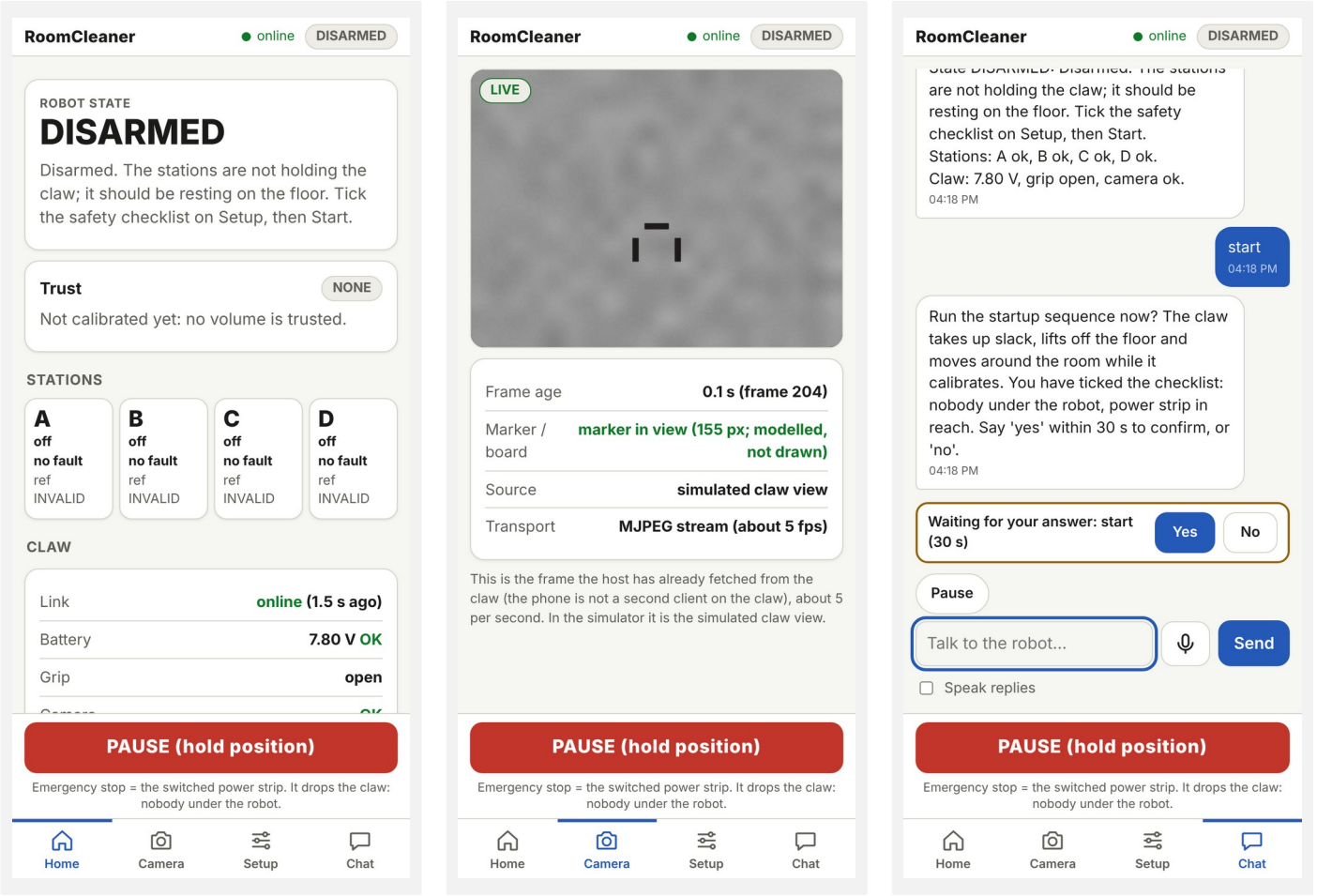
4. Put away

- Lower over the floor board
- Camera confirms the floor
- Disarm after a typed confirmation

- After each delivery the robot returns to step 2; three failures in a row pause it for the operator
- PAUSE or any fault stops every station at the same point on the path, still holding; nothing restarts by itself

Phone App

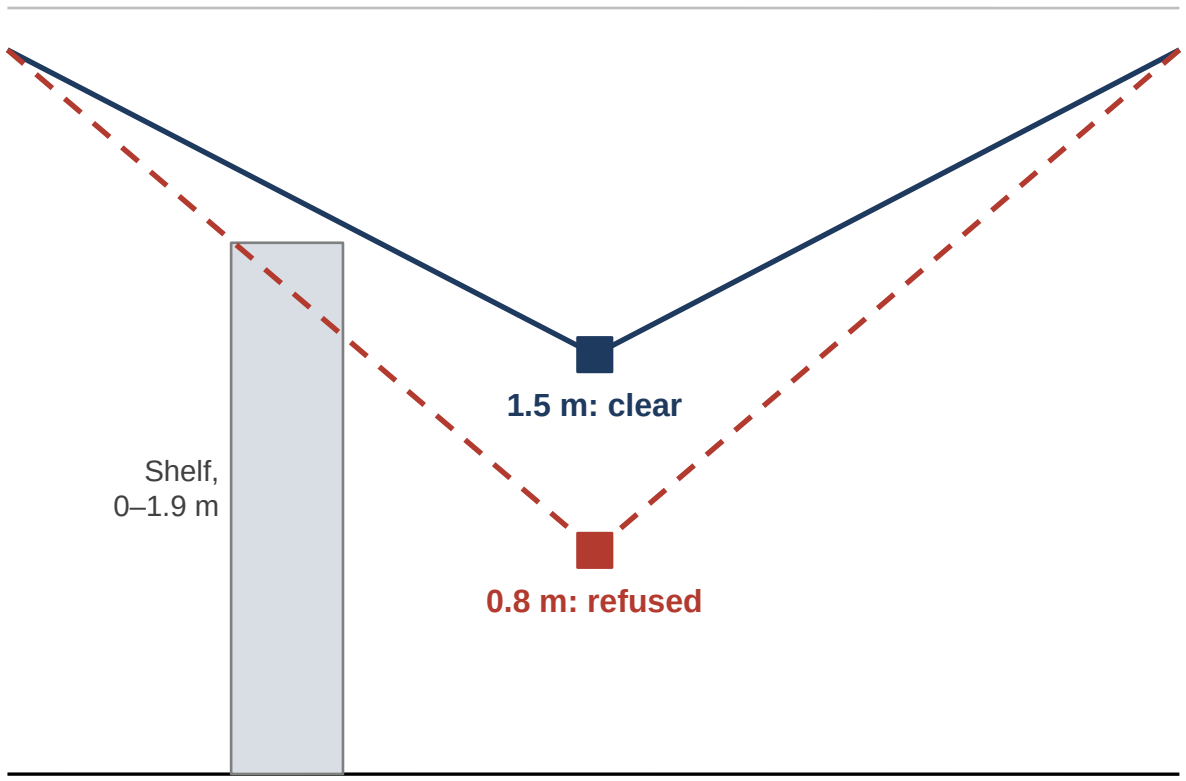
- Shows status, the live claw camera and guided setup
- Every motion asks for confirmation first
- PAUSE holds position immediately
- The switched power strip is the hard stop
- A reboot never resumes a task



Phone app: home, camera and chat screens

Obstacles

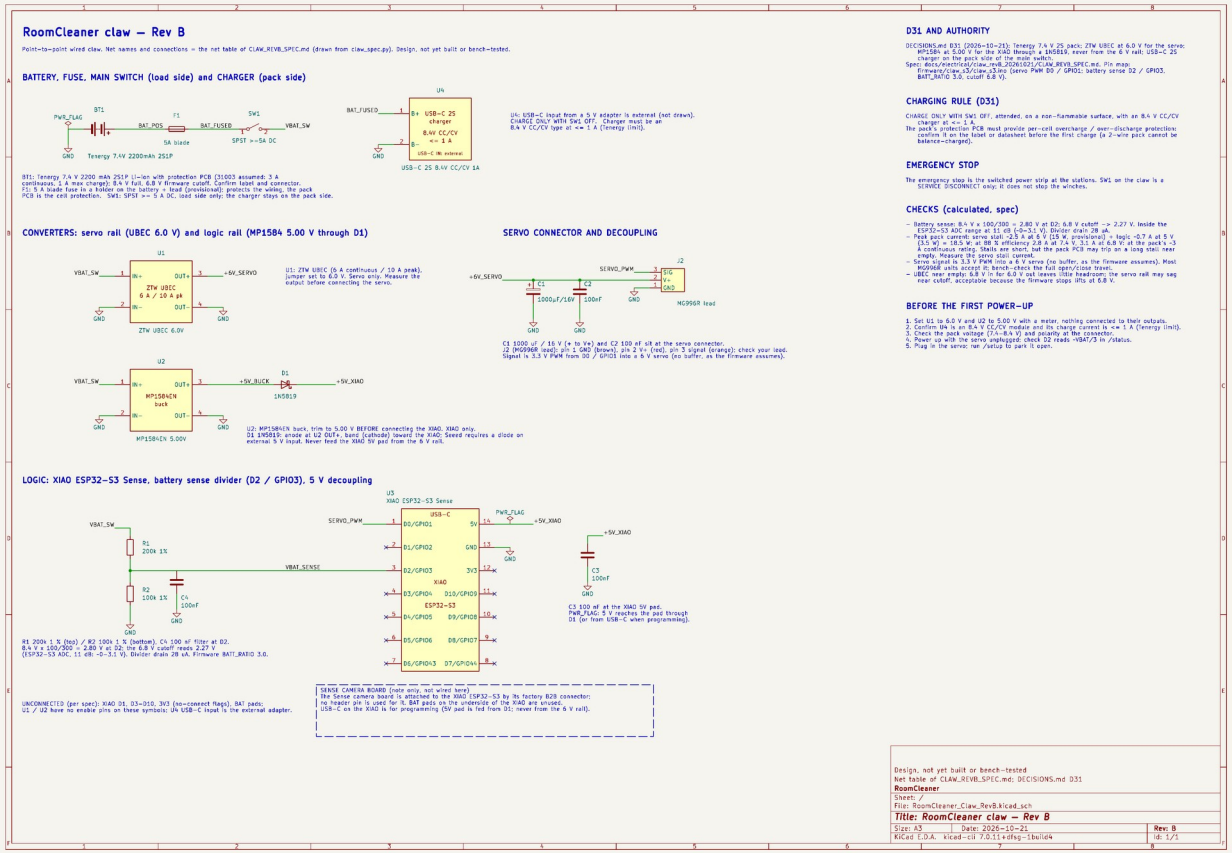
- Known obstacles modeled as keep-out cylinders
- Every cable checked at every point of the path
- Moves that fail the check are refused before any motor turns
- People are not detected yet, so nobody stands underneath



Side view along a room diagonal: the same floor point at two heights, with an example shelf

Claw Electrical

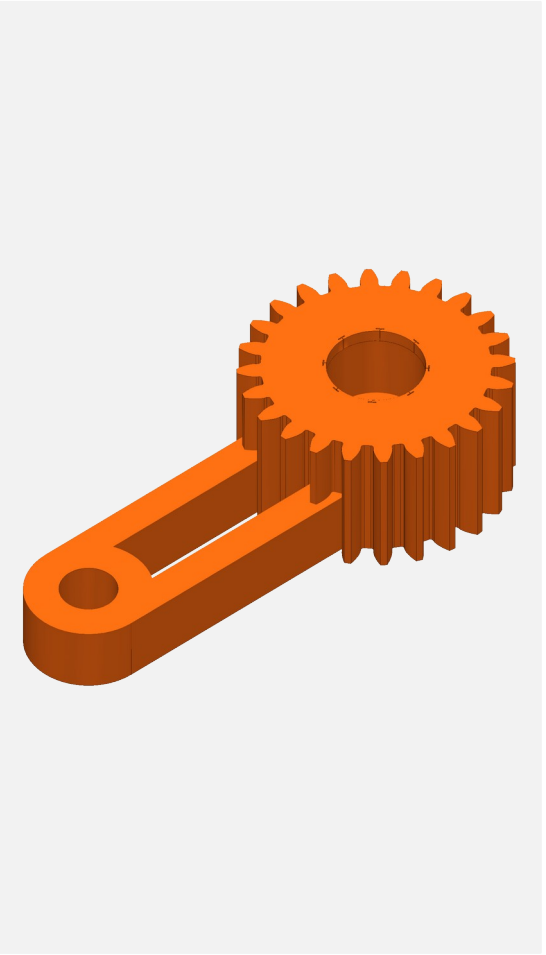
- Two-cell lithium-ion pack: 7.4 V with a 5 A fuse
- 6 V servo rail and a separate 5 V logic rail
- Peak draw of 2.8–3.1 A against a 3 A pack rating
- Charged with the switch off, attended, at 1 A or less



Gear Strength

- Module 1.5 gears; PETG allowable stress 22.5 MPa
- Safety factors at 6 V stall: 2.05 pinion, 2.50 compound gear, 2.00 sync gear
- Module 1 gave safety factors of 0.35–0.41 and was rejected





Lewis bending stress at servo stall

$$F_t = 2\tau / d = 2 \times 1.078 / 0.027 = 79.9\ \text{N}$$

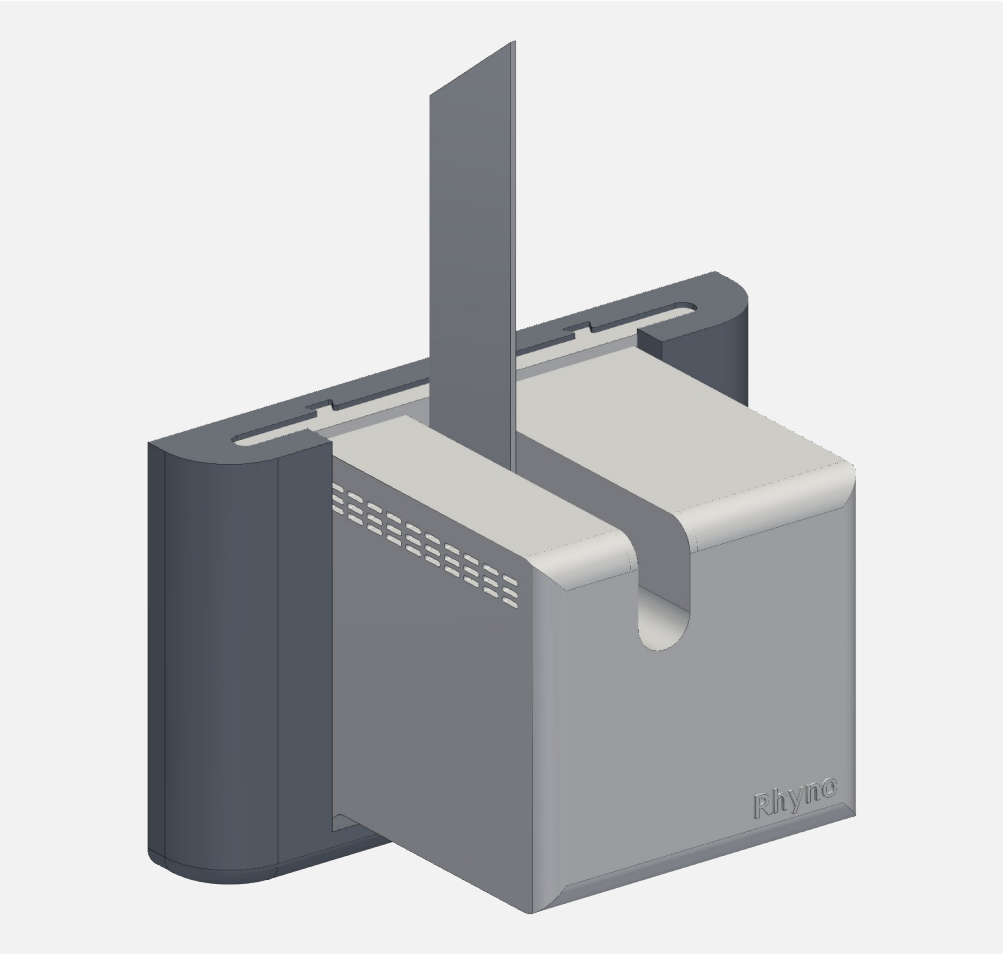
$$\sigma = F_t / (b\ m\ Y) = 79.9 / (15.7 \times 1.5 \times 0.309) = 11.0\ \text{MPa}$$

$$SF = \sigma_{allow} / \sigma = 22.5 / 11.0 = \mathbf{2.05}$$

Where: τ = servo stall torque; d = pitch diameter (18 teeth × 1.5 mm); b = face width; m = module; Y = Lewis form factor; SF = safety factor

Safety

- One hard stop: the switched power strip
- Faults and PAUSE hold motor torque
- Anchors in wall studs; nobody stands under the robot
- Motors are never unplugged under power

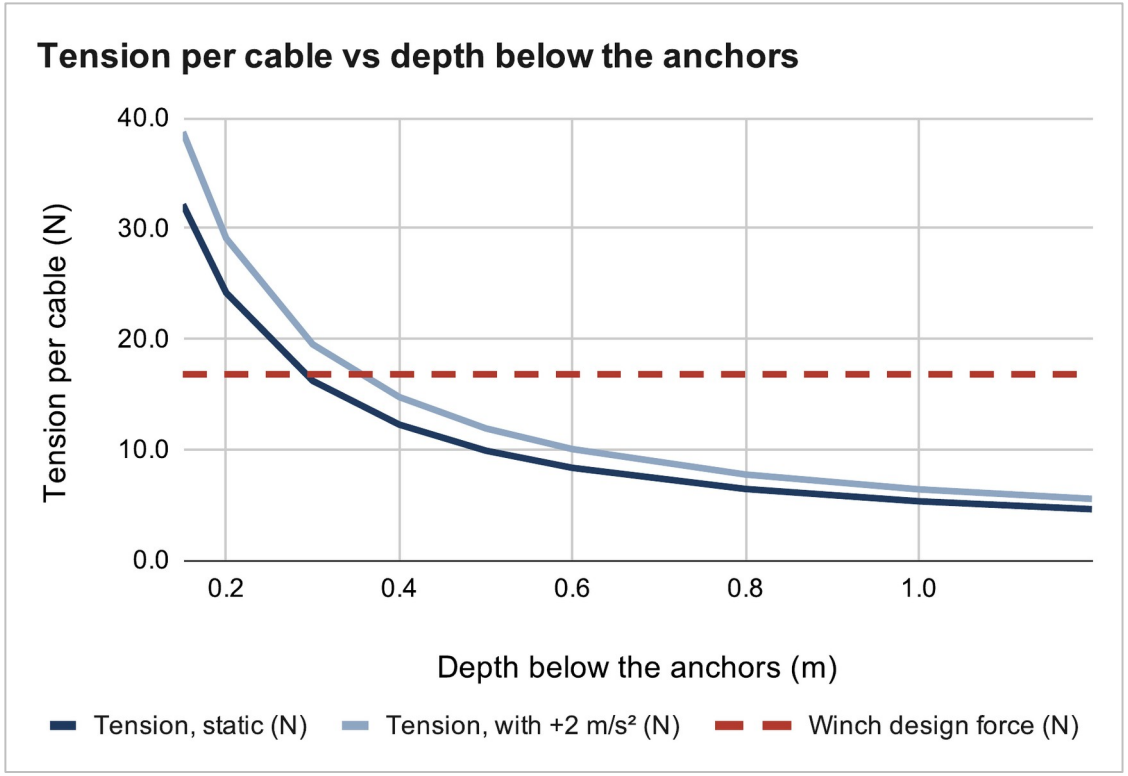


Acronyms

Term	Meaning
CAD	Computer-aided design
PETG	Polyethylene terephthalate glycol (the printed plastic)
TPU	Thermoplastic polyurethane (flexible printed plastic)
NEMA 17	National Electrical Manufacturers Association motor frame, 1.7 in square
I²C	Inter-Integrated Circuit: a short-range two-wire sensor bus
ESP32	A Wi-Fi microcontroller family (Espressif)
A4988, AS5600	Stepper-motor driver chip (Allegro); magnetic angle sensor (ams)
YOLO	“You Only Look Once”: an object-detection model family
SF	Safety factor: allowable stress divided by actual stress

Calculations and Sources

- All charts come from my engineering calculation sheet
- Each result is a formula on labeled inputs
- Sources:
  - Incropera et al., Fundamentals of Heat and Mass Transfer, ch. 3
  - Allegro MicroSystems, A4988 datasheet



Source: RoomCleaner engineering calculations, “Cable tension” (calculated values)

RoomCleaner engineering calculations (Google Sheets)